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Paper Code : PEC-CSBS- 501 B/PEC-IT501B Artificial Intelligence

UPID : 005510

Time Allotted : 3 Hours

Full Marks :70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

## Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[ 1 x 10 = 10 ]

- (I) An algorithm is \_\_\_\_\_ if it never overestimates the cost of reaching the goal.
- (II) Knowledge and reasoning also play a crucial role in dealing with \_\_\_\_\_ environment.
- (III) Resolution is also called single \_\_\_\_\_ rule.
- (IV) Choose from the following areas where NLP can not be useful.
  - A. Automatic Text Summarization
  - B. Medical transcription
  - C. Information Retrieval
  - D. Image Classification
- (V) In case a machine can change its course of action based on the external environment without any external help then the machine is called \_\_\_\_\_
- (VI) The problem-solving agent with several immediate options of unknown value can decide what to do by just examining different possible sequences of actions that lead to states of known value, and then choosing the best sequence. This process of looking for such a sequence is called \_\_\_\_\_.
- (VII) A rule-based system makes decision based on
  - A. set of constructs
  - B. set of predefined rules
  - C. set of clauses
  - D. set of predefined iterations
- (VIII) A system which is used to demonstrate, on a purely syntactic basis, that one formula is a logical consequence of another formula.
  - A. Inductive system
  - B. Deductive system
  - C. Abductive system
  - D. Intelligent system
- (IX) Which of the following is not an advantage of using an expert system development tool?
  - A. Imposed structure
  - B. Knowledge engineering assistance
  - C. Rapid prototyping
  - D. Human expertise
- (X) Artificial Intelligence is categorized into 3 steps :
  - A. Sensing, Reasoning, Acting
  - B. Collecting, Reasoning, Counting
  - C. Collecting, Repeating, Acting
  - D. Sensing, Repeating, Counting
- (XI) What is the evaluation function in A\* approach?
  - A. Cost from initial node to start node + Cost from start node to current node
  - B. Cost from initial node to start node + Cost from initial node to goal node
  - C. Cost from start node to current node + Cost from current node to goal node
  - D. Cost from current node to goal node + cost from initial node to goal node
- (XII) Which search is equal to minimax search but eliminates the branches that can't influence the final decision?
  - A. Gamma pruning
  - B. Alpha-beta pruning
  - C. Pre-train pruning
  - D. Post-train pruning

**Group-B (Short Answer Type Question)**

Answer any three of the following :

[ 5 x 3 = 15 ]

2. What is Artificial Intelligence? State and explain the concept of Turing Test in Artificial Intelligence. [5]
3. Explain A\* Algorithm with a sample graph example. [5]
4. Explain the knowledge cycle with a proper diagram. [5]
5. What is the disjunctive normal form? Find the disjunctive normal form for  $(p \rightarrow q) \wedge \neg r$  using truth table. [5]
6. (a) Explain the Advantages of Depth First Search and Breadth First Search. [5]  
(b) Explain OR graph with Example.

**Group-C (Long Answer Type Question)**

Answer any three of the following :

[ 15 x 3 = 45 ]

7. (a) Describe the generate and test algorithm with an example. [ 5+6+4 ]  
(b) Describe the four basic Measuring way to assess the performance of an algorithm  
(c) Explain Means-Ends Analysis with algorithm?
8. (a) Briefly describe the meaning of knowledge representation and knowledge acquisition. What procedure is followed for knowledge acquisition? Explain. [ 9 ]  
(b) Explain inference rules with examples. [ 6 ]
9. (a) Explain in detail about Resolution & Resolution inference Rule with example [ 7 ]  
(b) Define the syntactic elements of first-Order logic (ii) Illustrate the use of first-order logic to represent knowledge. <https://www.makaut.com> [ 8 ]
10. (a) Explain details about expert system shells and tools. [ 4+6+5 ]  
(b) Explain connectionist expert systems and inductive learning.  
(c) Explain selecting the appropriate problem for expert system.
11. (a) How categories are useful in knowledge representation? [ 5 ]  
(b) Explain the steps involved in the knowledge Engineering process. Give an example. Explain with an example. [ 10 ]

\*\*\* END OF PAPER \*\*\*